Transfer Learning for Covariance Matrix Estimation: Optimality and Adaptivity

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Abstract

Transfer learning, which leverages auxiliary data to improve inference in a target domain, has become a powerful tool in machine learning. In this work, we study minimax and adaptive estimation of large bandable covariance matrices under a transfer learning framework.

We first establish the minimax rate of convergence under squared spectral norm loss and propose a rate-optimal estimator. Our analysis reveals phase transitions determined by the source-target disparity, the source sample size, the dimension, and the two smoothness indices. We then develop an adaptive estimator, a fully data-driven procedure that does not require the unknown smoothness and disparity parameters. In most regimes, it directly attains the minimax optimal rate without additional cost of adaptation.

Simulation studies support our theoretical results and demonstrate the practical effectiveness of the proposed adaptive procedure. We further illustrate its utility in a phoneme classification task, where incorporating related phonemes as auxiliary data significantly improves classification performance.

Keywords: Adaptive covariance estimation, Covariance matrix, High-dimensional statistics, Minimax rate of convergence, Transfer learning.

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1 Introduction

In high-dimensional data analysis, understanding covariance structure is fundamental. Covariance matrices are central to regression, discriminant analysis, clustering, PCA, and graphical modeling. However, as dimensionality increases, accurate estimation becomes increasingly challenging, motivating extensive research into structured covariance models; for comprehensive reviews, see Cai et al. (2016); Cai (2017).

Among structured models, bandable covariance matrices—those with entries decaying away from the diagonal—have received significant attention due to their relevance in applications such as finance, biology, phonetics, climatology, and engineering (He and Chen, 2016; Qiu and Chen, 2012; Bien et al., 2016; Li et al., 2017; Lee et al., 2022; Qiu and Chen, 2015; Yi and Zou, 2013). Several estimation methods have been developed to exploit this structure, with minimax optimality (Bickel and Levina, 2008; Cai et al., 2010; Cai and Yuan, 2012).

Concurrently, *transfer learning*—which leverages information from auxiliary datasets to improve inference in a target domain—has become a powerful tool in modern machine learning. It has shown success in domains such as natural language processing, bioinformatics, medical imaging, and recommendation systems (Han et al., 2021; Petegrosso et al., 2017; Maqsood et al., 2019; Hu et al., 2019). For comprehensive surveys, see Weiss et al. (2016); Zhuang et al. (2021). Recent theoretical work has extended transfer learning to various statistical tasks, including nonparametric models, GLMs, multi-armed bandits, and functional estimation (Cai and Wei, 2021; Cai and Pu, 2022; Li et al., 2023; Cai, Cai and Li, 2024; Cai, Kim and Pu, 2024).

When transfer learning and covariance matrix estimation intersect, they create a synergy capable of addressing more complex and intriguing statistical problems. It enables more efficient use of related but distinct datasets, particularly when target data is lim-

ited or prohibitively expensive to collect. This synergy has proven useful in applications such as GTEx analysis, EEG-based BCI, and text classification (Li et al., 2022; Rodrigues et al., 2019; Raina et al., 2006), where transfer learning methods can substantially improve accuracy of covariance estimation.

1.1 Formulation of the Problem

Our discussion begins with an illustrative example of covariance matrix estimation within the transfer learning context, based on phoneme sound analysis. Phonemes—the fundamental units of sound in language—are typically represented by periodogram vectors, which capture sound intensity across a wide range of frequencies from short audio recordings of the corresponding phoneme (Hamooni and Mueen, 2014).

Zooming in on a particular phoneme in English, say `ZH` as in "seizure", we consider a target sample $\{X_1^{(t)}, \ldots, X_{n_t}^{(t)}\}$ where each observation $X_i^{(t)} \in \mathbb{R}^d$ is independently and identically distributed. Within the context of our phoneme study, the dimensionality d corresponds to the number of frequencies chosen to extract periodogram vectors from the audio recordings. Our goal is to estimate the target covariance matrix $\Sigma^{(t)} \in \mathbb{R}^{d\times d}$ of the periodogram vectors for the `ZH` phoneme. This estimation problem naturally arises in applications such as phoneme classification tasks (Hastie et al., 1995; Bien et al., 2016).

A key characteristic of the periodogram vector is its inherent ordering from low to high frequencies, which induces a distinctive structure in the target covariance matrix $\Sigma^{(t)}$. Entries close to the diagonal reflect covariances between intensities at nearby frequencies and are expected to exhibit larger magnitudes. In contrast, entries farther from the diagonal—corresponding to covariances between intensities at widely separated frequencies—tend to decay toward zero. This structure is clearly visualized in Figure 5 in Section 4.

The described structure of the phoneme dataset aligns with a bandable structure, a concept formulated by Bickel and Levina (2008). We examine the away-from-diagonal

operator, denoted $\mathfrak{A}_d$, on the set of $d \times d$ matrices. Given a bandwidth $u > 0$, the operator is defined by:

$$\mathfrak{A}_d(B; u) \coloneqq \Big(b_{jk} \mathbb{1}\big(|j-k| > u\big) \Big)_{j,k \in [d]} \qquad \text{where } B = (b_{jk})_{j,k \in [d]} \in \mathbb{R}^{d \times d}.$$

Figure 2a in Section 2 provides a visual representation of the away-from-diagonal operator $\mathfrak{A}_d$. It retains elements beyond a certain distance from the main diagonal. The bandwidth $u > 0$ given in the operator specifies this threshold of distance. Under the bandable structure, it is assumed that the magnitude of the away-from-diagonal part diminishes geometrically as the bandwidth increases. We define the class $\mathcal{B}_\alpha(M)$ for bandable covariance matrices characterized by a decay rate $\alpha > 0$ and radius $M > 0$ as follows:

$$\mathcal{B}_\alpha(M) \coloneqq \big\{ \Sigma \in \mathbb{S}_d^+ : \|\mathfrak{A}_d(\Sigma; u)\|_1 \leq M u^{-\alpha} \ \text{ for any } u > 0 \big\},$$

where $\mathbb{S}_d^+$ represents the collection of $d \times d$ symmetric and positive definite matrices and $\|\cdot\|_1$ indicates the matrix ℓ_1-norm.

The conventional learning framework centers around finding an effective method to estimate the target covariance matrix based on the available target sample. However, we would encounter a significant challenge within the conventional learning framework, particularly when it comes to the `ZH` phoneme. Data points for this phoneme are typically derived from continuous speech recordings, such as the TIMIT dataset provided by the Linguistic Data Consortium (Garofolo et al., 1993). However, the `ZH` phoneme is the least frequently occurring sound in everyday English speech (Hamooni and Mueen, 2014; Hayden, 1950), making its recordings scarce and the collection of additional data prohibitively expensive.

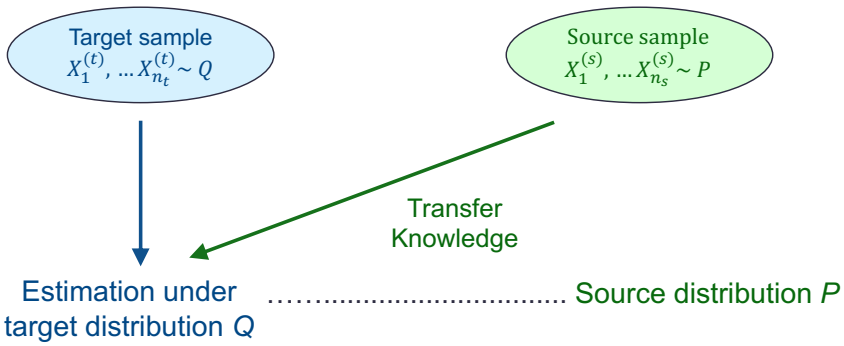


Figure 1: An illustration for transfer learning

One way to address this limitation is to extend the conventional learning framework through a transfer learning strategy. This approach allows us to utilize recordings of other phonemes that are different yet bear some similarities. For instance, the `S` phoneme, as in "see", is categorized as an obstruent fricative in the hierarchy of English phonemes, like the `ZH` phoneme (Dekel et al., 2004; Lee and Hon, 1989). Moreover, the `S` phoneme is significantly more common, ranking as the fifth most frequent sound in English speech (Hamooni and Mueen, 2014; Hayden, 1950). Consequently, we can leverage the similarity and larger sample size of the `S` phoneme to enhance the covariance matrix estimation for the `ZH` phoneme.

From a statistical perspective, in addition to having an independently and identically distributed target sample $\{X_1^{(t)}, \ldots, X_{n_t}^{(t)}\}$ from a target distribution Q, we have an auxiliary sample, $\{X_1^{(s)}, \ldots, X_{n_s}^{(s)}\}$, comprising d-dimensional random vectors that are independently and identically distributed from a source distribution P. The source sample is independent of the target sample and is also characterized by its source covariance matrix $\Sigma^{(s)} \in \mathbb{R}^{d \times d}$. While our objective remains the same, namely to find an optimal estimation method for the target covariance matrix $\Sigma^{(t)}$, our methodology may benefit from leveraging the source sample alongside the target sample. Figure 1 illustrates the transfer learning strategy.

The effectiveness of transfer learning depends on a meaningful and quantifiable rela-

tionship between the source and target samples. In this paper, we assume that:

- Both the target and source covariance matrices are bandable. Specifically, for target parameters (α_t, M_t) and source parameters (α_s, M_s), we assume $\Sigma^{(t)} \in \mathcal{B}_{\alpha_t}(M_t)$ and $\Sigma^{(s)} \in \mathcal{B}_{\alpha_s}(M_s)$.

- The disparity matrix $\Delta^{(s)} \coloneqq \Sigma^{(t)} - \Sigma^{(s)}$, which measures the deviation between the target and source covariance matrices, is sparse in the matrix ℓ_1-norm. For a given disparity threshold $\delta \geq 0$, we assume $\|\Delta^{(s)}\|_1 \leq \delta$.

More general disparity conditions and approximately bandable misspecification are discussed in the supplementary material.

Let $\mathcal{P}_{\alpha_t,\alpha_s,\delta}(M_t, M_s, \rho)$ denote the collection of joint distributions of the target and source samples satisfying these assumptions together with the uniform sub-Gaussian condition in Assumption 2.1 with parameter ρ for both samples. We focus on the bandable class as it serves as the canonical model for ordered variables and provides a transparent setting to study the interactions between structural decay, domain disparity, and relative sample sizes.

To establish minimax optimality, we evaluate estimators $\widehat{\Sigma}^{(t)}$ that utilize both target and source samples under the squared spectral-norm loss. The minimax risk is defined as:

$$\inf_{\widehat{\Sigma}^{(t)}} \sup_{\mathbb{P} \in \mathcal{P}_{\alpha_t,\alpha_s,\delta}(M_t,M_s,\rho)} \mathbb{E}\|\widehat{\Sigma}^{(t)} - \Sigma^{(t)}\|_2^2.$$

Our first main result characterizes the sharp rate of this risk.

1.2 Main Results and Our Contribution

Under the above model and loss, and for sufficiently large d, our first main result shows that the minimax risk satisfies the following rate whenever $\log d/n \to 0$ and $(\log d/n_t) \wedge \delta^2 \to 0$, the necessary and sufficient criterion for consistent estimation. By letting $n \coloneqq n_t + n_s$ and

$\alpha_{\max} \coloneqq \alpha_t \vee \alpha_s$, the minimax optimal rate is expressed as:

$$\Phi_{\alpha_t,\alpha_s,\delta}(n_t, n_s, d) \coloneqq \left(n^{-2\alpha_{\max}/(2\alpha_{\max}+1)} \wedge \frac{d}{n}\right) + \frac{\log d}{n} + \left(n_t^{-2\alpha_t/(2\alpha_t+1)} \wedge \frac{d}{n_t} \wedge \delta^2\right) + \left(\frac{\log d}{n_t} \wedge \delta^2\right). \tag{1}$$

That is, under the displayed consistency criterion, Equation (1) gives the order of the minimax risk for estimating the target covariance matrix in the transfer learning setting under the squared spectral-norm loss.

When there is no auxiliary source sample and no constraints on the disparity matrix, our setting simplifies to the conventional learning framework and the corresponding minimax rate $\Phi_{\alpha_t,\alpha_s,\infty}(n_t, 0, d)$ reduces to the known result in the conventional setting as detailed in Cai et al. (2010). Moreover, by comparing the minimax rate to the baseline performance, $\Phi_{\alpha_t,\alpha_s,\infty}(n_t, 0, d)$, we can identify the necessary and sufficient conditions under which the source sample and transfer learning are indeed effective. This identification reveals interesting phase transition phenomena.

The minimax optimal rate still leaves open the practical problem of constructing an estimator that does not depend on the unknown parameters α_t, α_s, or δ. The second major contribution of this paper is the *adaptive transfer block-tridiagonal estimator*, which does not require the smoothness or disparity parameters. In most of the regimes considered in this paper, this estimator directly attains the minimax optimal rate $\Phi_{\alpha_t,\alpha_s,\delta}(n_t, n_s, d)$, despite not knowing these parameters. This includes settings in which the source and target covariance matrices are sufficiently close, as well as broad regimes in which the source sample size and dimension grow regularly with the target sample size. The adaptive section describes these regimes and explains how the validation step governs the use of auxiliary data when transfer is less favorable.

We further demonstrate its practical effectiveness through both simulation studies and a real-data application. The simulation results validate the theoretical phase transition

phenomena: transfer learning improves estimation when the disparity is small, but offers relatively small benefit when the disparity is large. Additionally, we apply the adaptive estimator to a phoneme classification task using the TIMIT dataset. Incorporating auxiliary data from related phonemes substantially enhances classification accuracy in both QDA and LDA, with gains increasing as the auxiliary sample size grows, underscoring the practical utility of our proposed method.

1.3 Organization and Notation

The rest of the paper is organized as follows. We conclude this section by introducing the basic notation. Section 2 explores optimal transfer learning for covariance matrix estimation, presenting the blockwise tridiagonal estimator and establishing the minimax rate of convergence. Section 3 introduces the adaptive transfer estimator and discusses the regimes in which it attains Φ. Section 4 carries out simulation study and real-data analysis to evaluate the effectiveness of the proposed adaptive algorithm in various settings. Section 5 explores possible directions of future research. Finally, the proofs of the main theorems and technical results are presented in the supplementary material.

Throughout the paper, the primary asymptotic components are the sample sizes (n_t, n_s), the dimensionality (d) and the disparity threshold (δ), while all other variables are treated as constants. We conform to the conventional big-Oh(O), big-Omega(Ω) and big-Theta(Θ) notations. For conciseness, we occasionally substitute these notations with the symbols $\lesssim$, $\gtrsim$ and $\asymp$, respectively. Additionally, we adhere to the convention of little-oh(o) notation and simply write $a \ll b$ or $b \gg a$ to signify that $a = o(b)$.

In addition, let us introduce several notations pertinent to covariance matrices. We define the pooled covariance matrix $\Sigma \in \mathbb{R}^{d\times d}$, which represents the weighted average of the population covariance matrices, $\Sigma^{(t)}$ and $\Sigma^{(s)}$. Concurrently, we define another disparity matrix $\Delta \in \mathbb{R}^{d\times d}$ as the deviation of the target covariance matrix $\Sigma^{(t)}$ from the pooled

covariance matrix Σ. By letting $n \coloneqq n_t + n_s$, these are mathematically represented as:

$$\Sigma \coloneqq \frac{n_t}{n}\Sigma^{(t)} + \frac{n_s}{n}\Sigma^{(s)}, \qquad \text{and} \qquad \Delta \coloneqq \Sigma^{(t)} - \Sigma = \frac{n_s}{n}\Delta^{(s)}.$$

We proceed to define the sample counterparts of the covariance matrices. The target sample covariance matrix $\check{\Sigma}^{(t)}$ and the source sample covariance matrix $\check{\Sigma}^{(s)}$ are defined that for each group indicator $g \in \{t, s\}$,

$$\check{\Sigma}^{(g)} \coloneqq \frac{1}{n_g}\sum_{i=1}^{n_g}\big(X_i^{(g)} - \bar{X}^{(g)}\big)\big(X_i^{(g)} - \bar{X}^{(g)}\big)^\top \qquad \text{where } \bar{X}^{(g)} \coloneqq \frac{1}{n_g}\sum_{i=1}^{n_g} X_i^{(g)}.$$

Subsequently, we define the pooled sample covariance matrix $\check{\Sigma}$ as a weighted combination of the target and source sample covariance matrices:

$$\check{\Sigma} \coloneqq \frac{n_t}{n}\check{\Sigma}^{(t)} + \frac{n_s}{n}\check{\Sigma}^{(s)}.$$

Finally, it is apparent to define the sample disparity matrices $\check{\Delta}^{(s)}$ and $\check{\Delta}$ as:

$$\check{\Delta}^{(s)} \coloneqq \check{\Sigma}^{(t)} - \check{\Sigma}^{(s)}, \qquad \text{and} \qquad \check{\Delta} \coloneqq \check{\Sigma}^{(t)} - \check{\Sigma} = \frac{n_s}{n}\check{\Delta}^{(s)}.$$

2 Optimal Transfer Learning

In this section, our goal is to present an optimal method for estimating the target covariance matrix $\Sigma^{(t)}$ within the transfer learning framework. Recall that $\mathcal{P}_{\alpha_t,\alpha_s,\delta}(M_t, M_s, \rho)$ denotes the collection of joint distributions of the target and source samples such that $\Sigma^{(t)} \in \mathcal{B}_{\alpha_t}(M_t)$, $\Sigma^{(s)} \in \mathcal{B}_{\alpha_s}(M_s)$, $\|\Sigma^{(t)} - \Sigma^{(s)}\|_1 \le \delta$, and both samples satisfy the following uniform sub-Gaussianity condition:

Assumption 2.1 (Uniform sub-Gaussianity). The target observations $X_1^{(t)}, \ldots, X_{n_t}^{(t)}$ and the source observations $X_1^{(s)}, \ldots, X_{n_s}^{(s)}$ are uniformly sub-Gaussian random vectors with a standard-deviation proxy $\rho > 0$. More specifically, we assume:

$$\left.\begin{aligned} \big\|a^\top\big(X_1^{(t)} - \mathbb{E}X_1^{(t)}\big)\big\|_{\psi_2} \\ \big\|a^\top\big(X_1^{(s)} - \mathbb{E}X_1^{(s)}\big)\big\|_{\psi_2} \end{aligned}\right\} \le \rho, \qquad \text{for any unit vector } a \in \mathbb{R}^d,$$

where we denote by $\|\cdot\|_{\psi_2}$ the sub-Gaussian norm or Orlicz ψ_2-norm. As per Wainwright (2019) and Vershynin (2018), the sub-Gaussian norm of a random variable Z is defined by

$$\|Z\|_{\psi_2} := \inf\{r > 0 : \mathbb{E}e^{Z^2/r^2} \leq 2\}.$$

We are now ready to state the optimality theory for estimating the target covariance matrix under the transfer learning framework. The following theorem characterizes the minimax rate of convergence under the squared spectral-norm loss. In particular, it highlights how the interaction between the sample sizes (n_t, n_s), the dimension (d), and the disparity level (δ) governs the fundamental limits of transfer learning.

Theorem 2.2. *One can consistently estimate the target covariance matrix if and only if at least one of the following conditions holds:*

Assumption (a) $\log d \ll n_t$.

Assumption (b) $\log d \ll n$ *and* $\frac{\log d}{n_t} \wedge \delta^2 \ll 1$.

Under either condition, the optimal convergence rate under the squared spectral-norm loss is given by

$$\inf_{\widehat{\Sigma}^{(t)}} \sup_{\mathbb{P}\in\mathcal{P}_{\alpha_t,\alpha_s,\delta}} \mathbb{E}\|\widehat{\Sigma}^{(t)} - \Sigma^{(t)}\|_2^2 \asymp \left(n^{-2\alpha_{\max}/(2\alpha_{\max}+1)} \wedge \frac{d}{n}\right) + \frac{\log d}{n} + \left(n_t^{-2\alpha_t/(2\alpha_t+1)} \wedge \frac{d}{n_t} \wedge \delta^2\right) + \left(\frac{\log d}{n_t} \wedge \delta^2\right).$$

Remark 2.3. These results extend beyond the exact bandable model and the matrix ℓ_1-disparity condition. Under weaker assumptions, we replace the ℓ_1 constraint with an upper-tail stability (UTS) condition and allow for approximately bandable misspecification. In this broader setting, the same upper bound holds, supplemented only by a term quantifying the misspecification level.

We maintain the exact ℓ_1-based formulation in the main text to keep the phase-transition phenomena transparent. The more general statements are deferred to Subsection A.1 in

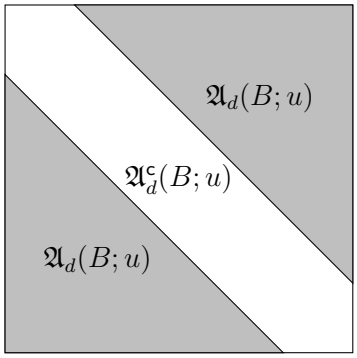
the supplementary material for pedagogical clarity, rather than any limitation in the theory itself. In the remainder of the paper, we retain the class $\mathcal{P}_{\alpha_t,\alpha_s,\delta}$ to streamline the presentation.

Due to space constraints, the proof of Theorem 2.2 is deferred to the supplementary material. We instead introduce the key technical ingredient for establishing the upper bound: the *blockwise tridiagonal operator*, denoted by $\mathfrak{T}_d$, which was previously employed in high-dimensional covariance estimation by Cai and Zhang (2016). This operator is defined for an integer bandwidth $u > 0$ and can be expressed as follows. Let $\lfloor x \rfloor$ denote the greatest integer not exceeding $x \in \mathbb{R}$.

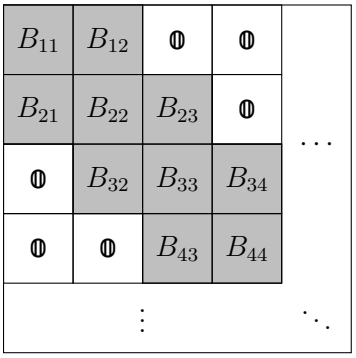
$$\mathfrak{T}_d(B; u) := \Big(b_{jk} \mathbb{1}\big(|u_j - u_k| \le 1\big) \Big)_{j,k\in[d]} \quad \text{where} \quad \begin{cases} B = (b_{jk})_{j,k\in[d]} \in \mathbb{R}^{d\times d}, \\ u_j := 1 + \big\lfloor (j-1)/u \big\rfloor, \ (j \in [d]). \end{cases}$$

Intuitively, when the matrix B is partitioned into $u \times u$ blocks, the operator $\mathfrak{T}_d$ retains only the blockwise main diagonal and the immediate super- and sub-diagonal blocks, resulting in a block-tridiagonal structure. A visual illustration of this mechanism is provided in Figure 2b.

By applying the blockwise tridiagonal operator $\mathfrak{T}_d$ with a carefully chosen bandwidth, we can establish the upper bound in the minimax rate of Theorem 2.2. Since the bandable structure implies decay of the away-from-diagonal parts, it is natural to truncate the sample covariance matrices using $\mathfrak{T}_d$. Although full sample covariance matrices perform poorly in high-dimensional settings, the low-dimensional blocks retained by $\mathfrak{T}_d$ can yield accurate local estimates, provided that the bandwidth is selected appropriately. Detailed derivations are provided in the supplementary material.



(a) The output, $\mathfrak{A}_d(B;u)$, of the away-from-diagonal operator



(b) The output, $\mathfrak{T}_d(B;u)$, of the blockwise tridiagonal operator

Figure 2: A visual representation of important matrix operators

Given the extensive literature on covariance matrix estimation under the conventional setting, it is instructive to compare our minimax result with known rates. Notably, the transfer learning framework reduces to the conventional framework when $n_s = 0$ and $\delta = \infty$, as the source sample is absent and the source population provides no information about the target covariance structure. In this case, Cai et al. (2010) established that the minimax rate of convergence is given by

$$\Phi_{\alpha_t,\alpha_s,\infty}(n_t, 0, d) \asymp \left(n_t^{-2\alpha_t/(2\alpha_t+1)} \wedge \frac{d}{n_t} \right) + \frac{\log d}{n_t}. \tag{2}$$

The optimal convergence rate established in Theorem 2.2 also provides insight into the effectiveness of the transfer learning. A natural question is to identify the conditions under which transfer learning yields tangible benefits, and to quantify the extent of such improvements. Notably, the analysis reveals a phase transition phenomenon, where the utility of transfer learning undergoes a sharp change depending on the disparity level (δ) and the source sample size (n_s). The main findings are concisely summarized in Table 1.

The disparity threshold δ plays a crucial role in the phase transition since it serves as a key measure of similarity between the target and source covariance matrices. Let us define

two critical points $\delta_1^* \lesssim \delta_2^*$ by

$$\delta_1^* := \left(n^{-\alpha_{\max}/(2\alpha_{\max}+1)} \wedge \sqrt{\frac{d}{n}} \right) + \sqrt{\frac{\log d}{n}}$$
$$\delta_2^* := \left(n_t^{-\alpha_t/(2\alpha_t+1)} \wedge \sqrt{\frac{d}{n_t}} \right) + \sqrt{\frac{\log d}{n_t}}.$$

Depending on the level of the disparity threshold δ, we naturally classify our model into three distinct categories: the minimal, the moderate and the strong disparity models as outlined in Table 1.

Recall that $\Phi^{\mathrm{LE}} := \Phi_{\alpha_t,\alpha_s,\infty}(n_t, 0, d)$ accords with the minimax rate of convergence in the conventional learning scenario. It is noteworthy that the minimax rate Φ must be no slower than Φ^{LE}, as it is possible to construct an estimator that relies solely on the target sample. Hence, the rate Φ^{LE} serves as a baseline and can be regarded as the least effective rate of convergence within the transfer learning framework.

Our model experiences a notable transition at the critical point $\delta \asymp \delta_2^*$, which dictates the potential benefits of transfer learning compared to conventional learning. Specifically, under the strong disparity model, characterized by $\delta \gtrsim \delta_2^*$, the minimax rate of convergence aligns exactly with Φ^{LE}. Despite the availability of the source sample, the disparity threshold δ is too substantial to facilitate the effective transfer of knowledge. This also highlights a safety property at the minimax level: as the disparity increases into the strong-disparity regime, the transfer benchmark degrades back to the target-only minimax rate Φ^{LE}, rather than becoming worse than the conventional baseline. Conversely, in the context of the minimal or moderate disparity model, characterized by $\delta \ll \delta_2^*$, incorporating the source sample presents an opportunity to enhance the accuracy of estimating the target covariance matrix.

Given that the disparity threshold is sufficiently small as in the minimal or moderate disparity models, the size of the source sample becomes a key factor in this enhancement. We explore the necessary and sufficient conditions for the enhancement in two separate

Table 1: Does transfer learning offer benefits? If so, does increasing the source sample size offer benefits?

<table>
<tr><th colspan="2"></th><th>Minimal disparity model ($\delta \ll \delta_1^*$)</th><th>Moderate disparity model ($\delta \gtrsim \delta_1^*$ and $\delta \ll \delta_2^*$)</th><th>Strong disparity model ($\delta \gtrsim \delta_2^*$)</th></tr>
<tr><td rowspan="2">Limited source sampling regime ($n_s \lesssim n_t$)</td><td>without auto-smoothing benefit</td><td colspan="2">No</td><td rowspan="2">No</td></tr>
<tr><td>with auto-smoothing benefit</td><td colspan="2"></td></tr>
<tr><td>Ample source sampling regime ($n_s \gg n_t$)</td><td></td><td>Yes / Yes</td><td>Yes / No</td><td></td></tr>
</table>

scenarios: the ample and the limited source sampling regimes as presented in Table 1. Additionally, a comparative analysis of these regimes reveals another phase transition phenomenon.

Within the ample source sampling regime where $n_s \gg n_t$ holds, it is readily verifiable that the minimax rate of convergence diminishes more rapidly than the baseline rate Φ^{LE}. In other words, a substantial size of the source sample is conducive to effective transfer learning, aligning with our expectations under the minimal or moderate disparity model. On the contrary, under the limited source sampling regime, defined by $n_s \lesssim n_t$, the effectiveness of the source sample may be compromised. Typically, the minimax rate of convergence aligns with the baseline rate Φ^{LE}. Nevertheless, the transfer learning can still benefit from the scarce source sample if the following extra conditions are satisfied:

- The source covariance matrix exhibits an expedited decay, as indicated by $\alpha_s > \alpha_t$.
- The dimensionality is neither too small nor large, as $n_t^{1/(2\alpha_s+1)} \ll d \ll \exp(n_t^{1/(2\alpha_t+1)})$.

These conditions give rise to the *auto-smoothing* effect, which allows the target covariance matrix to be analyzed as if it possesses the source decay rate α_s rather than the target decay rate α_t. If the source covariance matrix exhibits an expedited decay, the auto-smoothing now facilitates the more accurate estimation for the target covariance matrix. Auto-smoothing is akin to the concept of transferable smoothness described by Cai and Pu (2022) in the context of transfer learning for nonparametric regression. The dimensionality condition is equally critical, ensuring that the improved decay rate effectively aids in estimation. Therefore, those extra conditions are both essential and adequate to gain benefit from the auto-smoothing and to ensure effective transfer learning under the limited source sampling regime as outlined in Table 1.

Moving forward, our analysis now centers on regimes and models where transfer learning proves advantageous. Pertaining to the magnitude of this advantage, a final phase

transition is observed at the critical point, $\delta \asymp \delta_1^*$. Within the moderate disparity model, satisfying $\delta \gtrsim \delta_1^*$, the minimax rate of convergence is simply established as δ^2. Although the established rate indicates an enhancement over the baseline rate Φ^{LE}, the expansion of the source sample size fails to accelerate the convergence rate as long as other conditions remain the same. On the other hand, within the minimal disparity model, characterized by $\delta \ll \delta_1^*$, an increase in the source sample size yields additional benefits. It can be demonstrated that the minimax rate of convergence precisely matches the most efficient rate Φ^{ME}, defined as follows. By leveraging a model with a negligible disparity threshold, we can extract greater benefits from the larger source sample, leading to enhanced performance of estimation.

$$\Phi^{\mathrm{ME}} := \left(n^{-2\alpha_{\max}/(2\alpha_{\max}+1)} \wedge \frac{d}{n} \right) + \frac{\log d}{n}.$$

Let us further examine the case where the minimax rate coincides with the most effective rate Φ^{ME}. It is immediate that $\Phi^{\mathrm{ME}} = \Phi_{\alpha_{\max},\alpha_{\max},0}(n_t, n_s, d)$ holds, suggesting that the target and the source samples could be considered as originating from the same population. This representation is underscored by the auto-smoothing effect, which arises from the condition $\alpha_t = \alpha_s = \alpha_{\max}$. As we have discussed before, this effect equalizes the decay rates of the target and source covariance matrices, ensuring both exhibit the fastest decay rate, $\alpha_{\max}$. Not only that, another condition $\delta = 0$ indicates that the source and target covariance matrices are identical, effectively recasting the transfer learning framework into the conventional learning framework with augmented sample size, $n = n_t + n_s$. While the source population may differ from the target population, to estimate the target covariance matrix, the minimax rate of convergence guarantees that they are fundamentally equivalent.

3 Adaptive Transfer Learning

The minimax estimator in Section 2 uses the unknown smoothness indices α_t and α_s, as well as the unknown disparity level δ. We now construct a single data-driven estimator that does not use any of these parameters. The construction first fits an adaptive bandable covariance estimator to a single collection of observations, then compares its target-only and pooled instances through an independent one-dimensional validation statistic.

The benchmark is the ordinary classwise minimax rate $\Phi_{\alpha_t,\alpha_s,\delta}(n_t, n_s, d)$ in Equation (1). For later reference, let

$$p_m(a) \coloneqq m^{-2a/(2a+1)} \wedge \frac{d}{m}, \qquad P_m(a) \coloneqq p_m(a) + \frac{\log d}{m}, \qquad \alpha_{\max} \coloneqq \alpha_t \vee \alpha_s.$$

Then

$$\Phi_{\alpha_t,\alpha_s,\delta}(n_t, n_s, d) = P_n(\alpha_{\max}) + (p_{n_t}(\alpha_t) \wedge \delta^2) + \left(\frac{\log d}{n_t} \wedge \delta^2\right), \qquad n = n_t + n_s. \tag{3}$$

Our main result gives an additive oracle bound relative to this benchmark. Within the consistency regime of Theorem 2.2, it yields simultaneous minimax adaptivity up to a logarithmic factor, with exact matching when transfer is exact or sufficiently close to exact.

3.1 Adaptive transfer block-tridiagonal estimator

For convenience, we assume that $\mathbb{E}X_i^{(g)} = 0$ throughout this subsection for $g \in \{t, s\}$. For the general case, we may first apply the pairing symmetrization

$$W_i^{(g)} = \frac{X_{2i}^{(g)} - X_{2i-1}^{(g)}}{\sqrt{2}}, \qquad 1 \le i \le \left\lfloor \frac{N_g}{2} \right\rfloor, \qquad g \in \{t, s\},$$

to the original target and source samples, where N_g denotes the original sample size in group g. The paired observations are independent and satisfy $\mathbb{E}W_i^{(g)} = 0$, $\mathrm{Cov}(W_i^{(g)}) = \Sigma^{(g)}$, $\sup_{\|u\|_2=1} \|u^\top W_i^{(g)}\|_{\psi_2} \le C\rho$. In practice, one may instead center the samples by subtracting their empirical means, and the same theoretical results extend to this setting. Here and

below, let $L = C_L\rho^2$ be a fixed constant chosen so that the spectra of $\Sigma^{(t)}$ and $\Sigma^{(s)}$ lie in $[0, L]$.

A single-source adaptive estimator We first give a general single-source adaptive estimator as the building block of our adaptive transfer estimator. It treats the input data as a single sample, regardless of whether they come from the target sample or the pooled sample. The main idea is to construct a dyadic family of block-tridiagonal candidates, measure the cumulative spectral change from each candidate to its coarser scales, and select the bandwidth that minimizes this change together with a scale-dependent stochastic penalty. For a symmetric matrix B, we let $\mathsf{Clip}_L(B)$ denote the spectral clipping of B to $[0, L]$, obtained by truncating its eigenvalues in the spectral decomposition.

Algorithm 1 FitAdaptive

Require: A collection $\mathcal{D}$ of r independent observations; fixed constants λ and L.

1: Set $\widehat{\Sigma}_{\mathcal{D}} = r^{-1}\sum_{X\in\mathcal{D}} XX^\top$.

2: Set $u_- = d \wedge \lceil \log(ed) \rceil$, $u_+ = u_- \vee (d \wedge \lfloor r/\log(er) \rfloor)$, and $\mathcal{U}_r = \{2^k u_- : k \geq 0,\ 2^k u_- \leq u_+\}$.

3: **for** $u \in \mathcal{U}_r$ **do**

4: Set $\ell_u = \max\{\ell \geq 0 : 2^\ell u \leq u_+\}$ and

$$F_{\mathcal{D}}(u) = \sum_{\ell=1}^{\ell_u} \|\mathfrak{T}_d(\widehat{\Sigma}_{\mathcal{D}}; 2^\ell u) - \mathfrak{T}_d(\widehat{\Sigma}_{\mathcal{D}}; 2^{\ell-1}u)\|_2 + \lambda\rho^2\sqrt{\frac{u + \log(ed)}{r}}.$$

5: Set $\widehat{u} = \arg\min_{u\in\mathcal{U}_r} F_{\mathcal{D}}(u)$.

6: **return** $\mathsf{Clip}_L(\mathfrak{T}_d(\widehat{\Sigma}_{\mathcal{D}}; \widehat{u}))$.

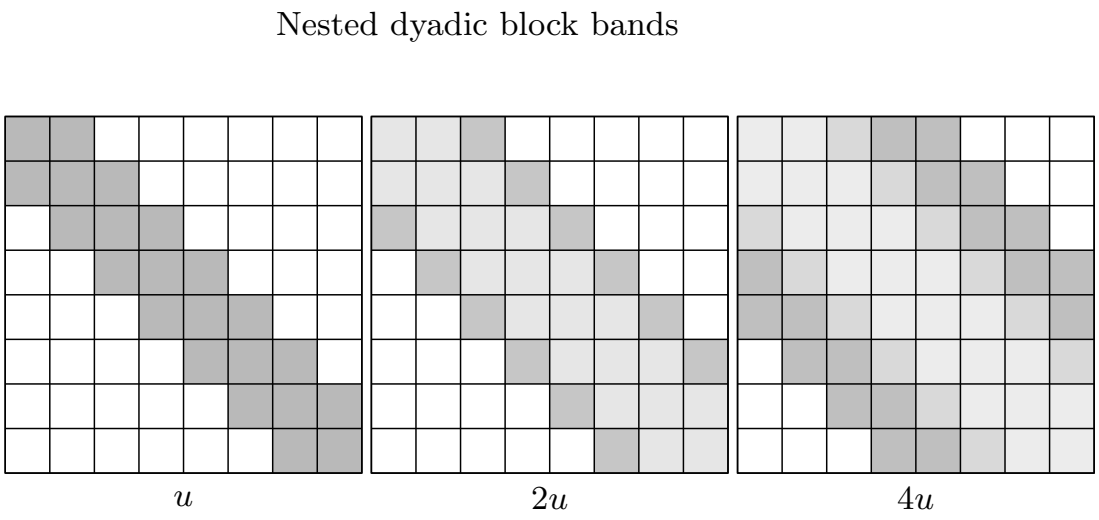


Figure 3: Nested dyadic block bands at scales u, $2u$, and $4u$.

Adaptive transfer estimator via validation For the adaptive transfer estimator, we first compute two candidates from the target samples only and the pooled samples, using the adaptive estimator in Algorithm 1. Then, we use a separate validation set from the target samples to choose between the two candidates. We use the estimator based on the pooled samples unless it deviates too much from the estimator based only on the target samples, in which case we fall back to the latter. The challenge is that directly validating two covariance matrices in high dimension with the operator norm incurs a dimension-dependent error, which can overwhelm the difference between the candidates. To avoid this cost, we first find a unit direction along which the two candidates differ most and then validate their projected quadratic forms in that direction using the held-out target samples. The validation problem is therefore scalar, so its statistical and computational cost is small. Moreover, when the two candidates are close in operator norm, either choice has essentially the same risk. The one-sided rule falls back to the estimator based only on the target samples when the validation data confirm that the candidate discrepancy is large enough to matter. This allows the estimator to retain the pooled candidate whenever transfer

is useful while protecting against substantial discrepancy between source and target, and hence to attain the minimax transfer rate without knowing the disparity level in advance.

Algorithm 2 Adaptive transfer block-tridiagonal estimator

Require: Independent target and source samples; a validation size $m_f \le n_t/2$; fixed constants λ, L, and C_τ.

Partition the target indices into disjoint $\mathcal{I}_t^{(0)}$ and $\mathcal{I}_t^{(f)}$, with $|\mathcal{I}_t^{(f)}| = m_f$ and $|\mathcal{I}_t^{(0)}| = n_t - m_f$.

Set $\mathcal{D}_t = \{X_i^{(t)} : i \in \mathcal{I}_t^{(0)}\}$, $\mathcal{D}_f = \{X_i^{(t)} : i \in \mathcal{I}_t^{(f)}\}$, and $\mathcal{D}_p = \mathcal{D}_t \cup \{X_i^{(s)} : 1 \le i \le n_s\}$.

Set $A_{\mathrm{tar}} = \mathsf{FitAdaptive}(\mathcal{D}_t)$ and $A_{\mathrm{pool}} = \mathsf{FitAdaptive}(\mathcal{D}_p)$.

Choose a unit v with $|v^\top (A_{\mathrm{tar}} - A_{\mathrm{pool}})v| = \|A_{\mathrm{tar}} - A_{\mathrm{pool}}\|_2$.

Set $\widehat{q} = \Pi_{[0,L]}\big(|\mathcal{D}_f|^{-1} \sum_{X \in \mathcal{D}_f} (v^\top X)^2\big)$, $a = v^\top A_{\mathrm{tar}} v$, and $b = v^\top A_{\mathrm{pool}} v$.

Set $\tau = C_\tau \rho^2 (\sqrt{\log(en)/m_f} + \log(en)/m_f)$.

if $|a - \widehat{q}| + 4\tau \le |b - \widehat{q}|$ **then**

    return $\widehat{\Sigma}_{\mathrm{ad}}^{(t)} = A_{\mathrm{tar}}$.

else

    return $\widehat{\Sigma}_{\mathrm{ad}}^{(t)} = A_{\mathrm{pool}}$.

3.2 Adaptive Upper Bound

We first provide the risk guarantees of the two candidate estimators over the target and pooled samples. These bounds separate the estimation error of the adaptive covariance fit from the discrepancy introduced by pooling.

Proposition 3.1 (Risk bounds for the two candidates)**.** *Suppose Assumption 2.1 holds and choose λ and C_L sufficiently large. Then the two calls to* FitAdaptive *satisfy*

$$\mathbb{E}\|A_{\mathrm{tar}} - \Sigma^{(t)}\|_2^2 \lesssim P_{n_t}(\alpha_t), \tag{4}$$

$$\mathbb{E}\|A_{\mathrm{pool}} - \Sigma^{(t)}\|_2^2 \lesssim P_n(\alpha_{\max}) + \delta^2. \tag{5}$$

The target candidate therefore achieves $P_{n_t}(\alpha_t)$, the optimal rate for estimating $\Sigma^{(t)}$ from the target sample alone. The pooled candidate achieves $P_n(\alpha_{\max}) + \delta^2$, which is the

rate obtained by treating the target and source observations as a single pooled sample, up to the squared discrepancy between their covariance matrices. These are exactly the two candidate rates identified in Section 2.

With these candidate bounds in hand, the remaining task is to show that the directional validation rule selects a candidate with the appropriate risk. The following theorem establishes this oracle behavior without requiring prior knowledge of either smoothness index or the disparity level. Define the general selector remainder

$$\Gamma_{m_f,n} \coloneqq \rho^4 \left(\frac{\log(en)}{m_f} + \frac{\log^2(en)}{m_f^2} \right). \tag{6}$$

Theorem 3.2. *Suppose Assumption 2.1 holds and choose λ, C_L, C_τ sufficiently large. For every fixed $\alpha_t, \alpha_s > 0$ and every $\delta \geq 0$, the estimator in Algorithm 2 satisfies*

$$\sup_{\mathbb{P} \in \mathcal{P}_{\alpha_t,\alpha_s,\delta}} \mathbb{E} \|\widehat{\Sigma}_{\mathrm{ad}}^{(t)} - \Sigma^{(t)}\|_2^2 \lesssim \Phi_{\alpha_t,\alpha_s,\delta}(n_t, n_s, d) + (\delta^2 \wedge \Gamma_{m_f,n}). \tag{7}$$

The implicit constant is uniform when (α_t, α_s) ranges over a fixed compact subset of $(0,\infty)^2$.

Theorem 3.2 shows that adaptively selecting between the two candidates only suffers from an additive cost $\delta^2 \wedge \Gamma_{m_f,n}$. In many cases, the additive cost can be absorbed into the optimal rate $\Phi_{\alpha_t,\alpha_s,\delta}(n_t, n_s, d)$, which we recall is defined in Equation (1).

Corollary 3.3. *Under the conditions of Theorem 3.2, suppose either of the following holds:*

- $\delta^2 \lesssim p_{n_t}(\alpha_t) \vee \frac{\log d}{n_t}$;
- *for fixed $\alpha_t, \alpha_s > 0$, $m_f = \lfloor n_t / \log(en_t) \rfloor$, $n_s \asymp n_t^\kappa$, and $d \asymp n_t^\xi$, with $\kappa \geq 0$ and $\xi > 0$.*

Then

$$\sup_{\mathbb{P} \in \mathcal{P}_{\alpha_t,\alpha_s,\delta}} \mathbb{E} \|\widehat{\Sigma}_{\mathrm{ad}}^{(t)} - \Sigma^{(t)}\|_2^2 \lesssim \Phi_{\alpha_t,\alpha_s,\delta}(n_t, n_s, d). \tag{8}$$

Our theory shows that adaptation can still be achieved without an additional leading-order cost under transfer learning. In the two particular settings in Corollary 3.3, the

adaptivity is achieved without additional cost. First, a small disparity is enough to make the adaptivity cost disappear: δ^2 only needs to be no larger than the statistical scale for estimating from target data alone, namely $p_{n_t}(\alpha_t) \vee (\log d)/n_t$, which includes exact transfer as a special case. Second, the adaptivity cost also disappears under a deliberately mild polynomial growth condition. It allows both the source sample size and the dimension to grow polynomially with the target sample size, with $n_s \asymp n_t^{\kappa}$, $d \asymp n_t^{\xi}$, and a validation size of order $n_t/\log n_t$. It imposes no small disparity condition and no ordering condition between α_t and α_s. Outside these two regimes, Theorem 3.2 isolates the price of transfer adaptation in the additive term $\delta^2 \wedge \Gamma_{m_f,n}$, rather than a multiplicative factor. This term is generally small when the validation size m_f is chosen proportional to n_t.

The low cost of adaptivity comes from our validation strategy, which reduces the comparison of the two covariance candidates to a scalar validation problem. We choose a direction that exposes the largest discrepancy between the candidates and compare their projected quadratic forms along this direction, rather than estimating an operator norm over the full dimension. The resulting concentration cost depends on the validation size and the confidence level, but does not introduce an additional factor that grows with d. This separation is important in high dimensions, where a direct operator norm comparison could obscure the difference between the candidates.

The procedure also has a natural safety interpretation. When the pooled candidate is compatible with the target covariance, the validation step retains it and makes use of the larger effective sample and the smoother covariance structure available after pooling. When the discrepancy is too large for the pooled candidate to be reliable, the one-sided rule can fall back to the candidate built from the target data alone. The same estimator therefore adapts simultaneously to the two smoothness indices, the source sample size, and the unknown transfer quality, while keeping the candidate built from target data alone available as a safety benchmark.

In this sense, the phase transitions described in Section 2 characterize when transfer is beneficial rather than how the estimator must be tuned. In practice, our adaptive procedure does not need to classify the problem into a strong, moderate, or minimal disparity regime, or into an ample or limited source regime, before fitting the estimator. Those regimes emerge only in the risk interpretation of the same adaptive procedure.

4 Numerical Experiments

This section evaluates the numerical performance of the adaptive transfer estimator, as depicted in Theorem 3.2. Subsection 4.1 examines its behavior under varying source sample sizes (n_s) and disparity levels (δ), confirming consistency with our theoretical findings. Subsection 4.2 demonstrates its practical utility in a phoneme classification task. Supplementary Material also contains additional experiments that examine robustness beyond the baseline setting, including more general source-target disparity structures, misspecification of exact bandability, and non-Gaussian sampling.

4.1 Simulation Study

We consider the target and source covariance matrices as follows:

$$\Sigma^{(t)} := \left(\sigma_{jk}^{(t)}\right)_{j,k\in[d]} \qquad \text{where } \sigma_{jk}^{(t)} := \begin{cases} 1 & \text{if } j = k, \\ C_t|j-k|^{-3}\xi_{jk}^{(t)} & \text{if } j \neq k, \end{cases}$$

$$\Sigma^{(s)} := \left(\sigma_{jk}^{(s)}\right)_{j,k\in[d]} \qquad \text{where } \sigma_{jk}^{(s)} := \begin{cases} 1 & \text{if } j = k, \\ \sigma_{jk}^{(t)} + C_s|j-k|^{-2}\xi_{jk}^{(s)} & \text{if } j \neq k. \end{cases}$$

In this context, $\xi_{jk}^{(t)}$ and $\xi_{jk}^{(s)}$ $(1 \leq j < k \leq d)$ represent independent Rademacher variables with $\xi_{jk}^{(t)} = \xi_{kj}^{(t)}$ and $\xi_{jk}^{(s)} = \xi_{kj}^{(s)}$ $(j, k \in [d])$. The constant C_t is set at 0.1, while the other constant $C_s > 0$ is adjusted in accordance with the disparity threshold $\delta := \|\Sigma^{(t)} - \Sigma^{(s)}\|_1$.

Notably, the target covariance matrix $\Sigma^{(t)}$ exhibits a target decay rate of $\alpha_t = 2$, whereas the source covariance matrix $\Sigma^{(s)}$ presents a source decay rate of $\alpha_s = 1$. This configuration merits careful attention because the condition $\alpha_s < \alpha_t$ makes the source covariance less regular than the target covariance and introduces additional challenges in estimating the target covariance matrix.

The target sample size n_t and the dimensionality d remain fixed at $(n_t, d) = (50, 50)$, reflecting their non-critical role in the effectiveness of transfer learning. Our investigation instead includes various combinations of the source sample size n_s and the disparity threshold δ. The source sample size is assessed at $n_s \in \{50, 100, 200, 400, 800\}$, and the disparity threshold is evaluated at $\delta \in \{0, 0.1, 0.4, 0.7, 1, 1.3\}$. Importantly, we introduce the scenario $(n_s, \delta) = (0, \infty)$ for comparative analysis. This scenario represents the conventional learning paradigm and thus establishes the baseline performance for our simulation study.

For each specified configuration, we repeat 1,000 simulations to generate target and source samples from mean-zero Gaussian distributions. Upon each simulated dataset, we employ the adaptive transfer estimator, as described in Theorem 3.2, and calculate its squared spectral-norm loss. The comprehensive results are visually represented in Figure 4, which unveils several intriguing patterns and corroborates our theoretical findings.

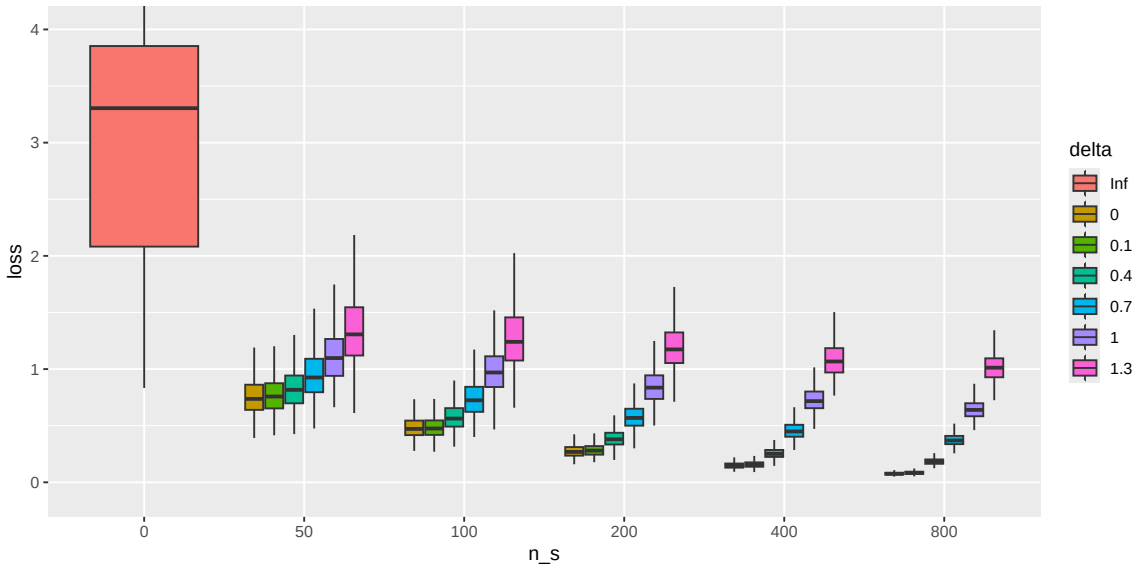


Figure 4: The results of numerical experiments under $\alpha_t = 2$ and $\alpha_s = 1$

First, when $\delta = 1.3$, transfer learning still improves over the baseline scenario, where $(n_s, \delta) = (0, \infty)$, although the gain is much smaller than in the low-disparity configurations. For small and moderate source sample sizes, the substantial discrepancy between the target and source covariance matrices limits the useful information to borrow. As n_s becomes larger, the loss falls further below the target-only baseline.

The configuration $\delta = 1.3$ represents a high-disparity setting in which borrowing information from the source is more challenging. Nevertheless, as depicted in Figure 4, the performance of the adaptive transfer estimator under $\delta = 1.3$ does not markedly deteriorate compared to the target-only baseline $(n_s, \delta) = (0, \infty)$. Thus, under the displayed settings, the adaptive transfer estimator remains competitive with the target-only baseline even when the source-target discrepancy is large.

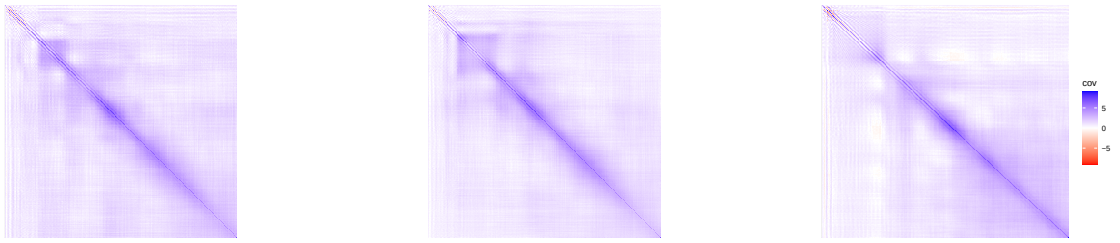
Conversely, when δ is set below 1.3, Figure 4 exhibits an evident improvement relative to the baseline scenario, where $(n_s, \delta) = (0, \infty)$. This improvement amplifies as the source sample size n_s expands or as the disparity threshold δ diminishes. Remarkably, the model's performance at $\delta = 0.1$ is nearly indistinguishable from that at $\delta = 0$, suggesting that the

source sample generated with $\delta = 0.1$ is as potent as one with $\delta = 0$. These empirical findings are consistent with our theoretical assertions and validate in practice the effectiveness of the adaptive transfer estimator.

4.2 Application to Discriminant Analysis of Phoneme Dataset

In this section, we apply the adaptive transfer estimator to a real-world dataset. Since the true covariance matrix is generally unknown in practice, directly assessing the accuracy of our estimate is infeasible. However, any method that relies on covariance estimation is expected to benefit from a more accurate covariance estimator when a similar but distinct source sample is available. To evaluate the practical utility of our estimator, we employ discriminant analysis in a classification problem, using classification accuracy as an indirect measure of performance. Specifically, we anticipate higher accuracy in discriminant analysis when incorporating a source sample and applying our covariance estimator, as knowledge transfer should lead to a more accurate estimate of the covariance matrix.

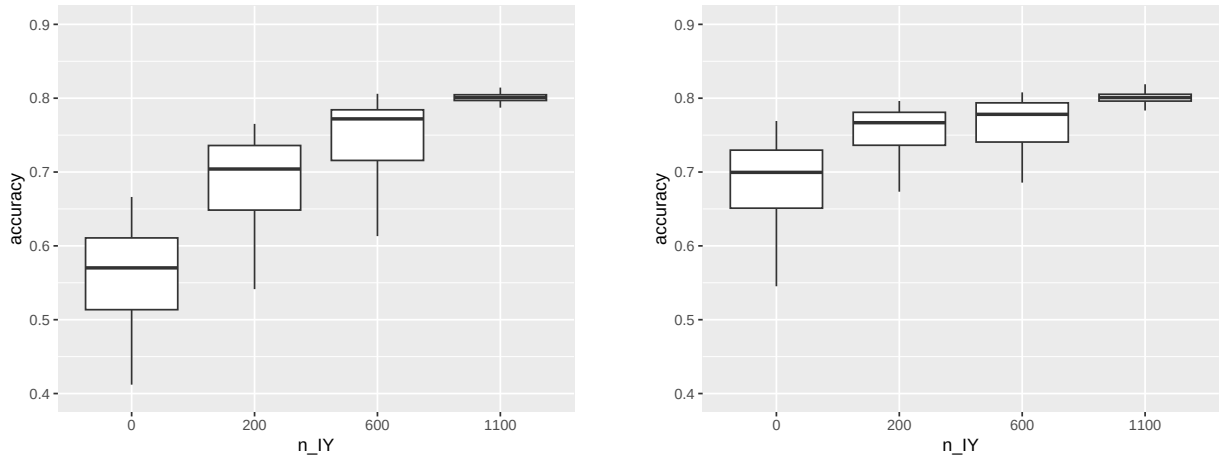
We consider a binary classification problem as described in Hastie et al. (2009). The dataset is derived from the TIMIT database, a widely used resource for speech recognition research, provided by the Linguistic Data Consortium (Garofolo et al., 1993). The processed phoneme benchmark considered here follows the version used in Hastie et al. (1995). It contains short sound recordings of male speakers pronouncing one of two similar-sounding phonemes. The objective is to develop a classifier for automatically labeling these sounds. The predictor vectors, x, are log-periodograms, which represent the log-intensity of the sound recordings across $d = 256$ frequencies. Each log-periodogram is labeled with its corresponding phoneme, y.



(a) AA (b) AO (c) IY

Figure 5: Sample Covariance Matrices of Three Phoneme Sounds

Our analysis focuses on three phonemes: AA as the vowel in "dark", AO as the first vowel in "water" and IY as the vowel in "she". Specifically, we develop a classifier to distinguish between AA and AO phonemes. They sound very similar and are difficult to differentiate, even for human listeners, given the short duration of the recordings. We randomly split the data into training and test sets, with the training set containing AA samples of $n_{\texttt{AA}} = 69$ and AO samples of $n_{\texttt{AO}} = 102$.



(a) QDA Classifiers (b) LDA Classifiers

Figure 6: Accuracy of Discriminant Analysis of Phoneme Dataset

Note that the training set is relatively small compared to the dimension $d = 256$, which hinders both the accurate estimation of the covariance matrix and the performance of discriminant analysis. One way to mitigate the effects of limited sample size is to integrate

similar but distinct sounds. In particular, the IY sound is classified as a sonorant vowel, like both AA and AO (Dekel et al., 2004; Lee and Hon, 1989). Figure 5 supports the use of the adaptive transfer estimator, as all covariance matrices exhibit a bandable structure, and IY shows some degree of similarity with both AA and AO.

With 1,163 auxiliary IY samples available, we utilize subsets of these to enhance covariance matrix estimation for Quadratic Discriminant Analysis (QDA) and Linear Discriminant Analysis (LDA). Specifically, we randomly sample IY sounds at sizes $n_{\texttt{IY}} \in \{0, 200, 600, 1100\}$. The case $n_{\texttt{IY}} = 0$ represents the target-only version of the adaptive transfer estimator, without auxiliary data.

For each random split, LDA utilizes the pooled sample covariance matrix of the AA and AO training data, while QDA employs class-specific sample covariance matrices. When $n_{\texttt{IY}} > 0$, we compute the sample covariance of the IY subsample and provide it as the source input to the adaptive transfer estimator—applied once for the pooled covariance in LDA and separately for the two class-specific matrices in QDA. Class means and priors are consistently estimated from the AA/AO training data alone. The resulting covariance estimates are then plugged into the standard LDA or QDA decision rules.

Theoretical support for this approach is provided in Proposition E.1 of Supplementary Material E, which links operator-norm covariance error to excess classification risk in binary Gaussian LDA. The classification accuracy of these models is presented in Figure 6. Across the 100 random splits, the mean accuracies at $n_{\texttt{IY}} \in \{0, 200, 600, 1100\}$ are $(0.562, 0.683, 0.741, 0.801)$ for QDA and $(0.677, 0.743, 0.752, 0.801)$ for LDA, respectively. Thus, on this grid, accuracy increases monotonically with the auxiliary sample size; at $n_{\texttt{IY}} = 1100$, the gains over the target-only fit are 0.239 for QDA and 0.124 for LDA.

5 Discussions

This paper explores minimax and adaptive estimation of high-dimensional bandable covariance matrices within a transfer learning framework. We establish the minimax rate of convergence under squared spectral-norm loss and introduce a blockwise tridiagonal estimator that achieves optimality. This minimax optimal rate, denoted by Φ, reveals phase transitions that characterize when source samples improve target-domain estimation.

For unknown smoothness and disparity parameters, we develop the adaptive transfer block-tridiagonal estimator. Its risk satisfies the global additive oracle bound $\Phi + \big(\delta^2 \wedge \Gamma_{m_f,n}\big)$, where the selector remainder depends on the validation size and only logarithmically on the total sample size. The remainder is absorbed by Φ under near transfer, including exact transfer, and under the stated polynomial growth regime for the source sample size and dimension.

The numerical results provide complementary evidence for the adaptive transfer estimator. Across the reported source-sample sizes and disparity levels, transfer gains are larger for smaller disparity and larger source sample sizes, whereas at high disparity the adaptive estimator remains close to the target-only baseline. Application to the phoneme data further demonstrates the usefulness of our adaptive estimator.

Our study focuses on estimation under the squared spectral-norm loss. While conventional bandable covariance estimation has been investigated under other measures—such as the matrix ℓ_1-norm, Frobenius norm, and general Bregman divergences—extending these to the transfer learning framework represents a compelling direction for future work. Similarly, extending these results to other structures, such as sparse and spiked covariance matrices, or to the estimation of bandable precision matrices, remains an open and promising area of research.

Finally, we identify robustness to heavy-tailed or contaminated observations as a critical

future direction. Although simulations in the supplementary material suggest the proposed procedure remains empirically competitive beyond Gaussian settings, a rigorous extension would require robust estimators tailored to the transfer setting. Potential avenues include matrix-truncation and spectrum-truncation approaches (Minsker and Wei, 2018; Ke et al., 2019), or contamination-robust procedures under Huber's model (Chen et al., 2018; Avella-Medina et al., 2018). Retaining sharp upper bounds and adaptive guarantees in these robust-transfer settings remains a substantial methodological challenge.

These topics are left for future investigation. We anticipate that the framework, insights, and theoretical results presented here will serve as foundational resources for subsequent studies in this domain.

SUPPLEMENTARY MATERIAL

Title: Supplement to "Transfer Learning for Covariance Matrix Estimation: Optimality and Adaptivity."

This supplementary material provides the complete proofs of the main theorems and technical results presented in the paper titled "Transfer Learning for Covariance Matrix Estimation: Optimality and Adaptivity." The contents are organized as follows. Supplementary Material A introduces fundamental notation, develops weaker-assumption and misspecified-model extensions through Proposition A.1 and Theorems A.3 and A.4, and contains the proofs of Theorems A.5, 2.2 and 3.2. Supplementary Material B and Supplementary Material C provide proofs of the technical results for optimality and adaptive aggregation, respectively. Supplementary Material D contains additional numerical experiments for the adaptive transfer estimator. Supplementary Material E gives a preliminary classification guarantee for plug-in LDA.

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References

Avella-Medina, M., Battey, H. S., Fan, J. and Li, Q. (2018), 'Robust estimation of high-dimensional covariance and precision matrices', *Biometrika* **105**(2), 271–284.

Bickel, P. J. and Levina, E. (2008), 'Regularized estimation of large covariance matrices', *The Annals of Statistics* **36**(1), 199–227.

Bien, J., Bunea, F. and Xiao, L. (2016), 'Convex Banding of the Covariance Matrix', *Journal of the American Statistical Association* **111**(514), 834–845.

Cai, C., Cai, T. T. and Li, H. (2024), 'Transfer learning for contextual multi-armed bandits', *The Annals of Statistics* **52**(1), 207–232.

Cai, T. T. (2017), 'Global Testing and Large-Scale Multiple Testing for High-Dimensional Covariance Structures', *Annual Review of Statistics and Its Application* **4**(Volume 4, 2017), 423–446.

Cai, T. T., Kim, D. and Pu, H. (2024), 'Transfer learning for functional mean estimation: Phase transition and adaptive algorithms', *The Annals of Statistics* **52**(2), 654–678.

Cai, T. T. and Pu, H. (2022), 'Transfer learning for nonparametric regression: Non-asymptotic minimax analysis and adaptive procedure', *Technical report* .

Cai, T. T., Ren, Z. and Zhou, H. H. (2016), 'Estimating structured high-dimensional covariance and precision matrices: Optimal rates and adaptive estimation', *Electronic Journal of Statistics* **10**(1), 1–59.

Cai, T. T. and Wei, H. (2021), 'Transfer Learning for Nonparametric Classification: Minimax Rate and Adaptive Classifier', *The Annals of Statistics* **49**(1), 100–128.

Cai, T. T. and Yuan, M. (2012), 'Adaptive covariance matrix estimation through block thresholding', *The Annals of Statistics* **40**(4), 2014–2042.

Cai, T. T. and Zhang, A. (2016), 'Minimax rate-optimal estimation of high-dimensional covariance matrices with incomplete data', *Journal of Multivariate Analysis* **150**, 55–74.

Cai, T. T., Zhang, C.-H. and Zhou, H. H. (2010), 'Optimal rates of convergence for covariance matrix estimation', *The Annals of Statistics* **38**(4), 2118–2144.

Chen, M., Gao, C. and Ren, Z. (2018), 'Robust covariance and scatter matrix estimation under huber's contamination model', *The Annals of Statistics* **46**(5).

Dekel, O., Keshet, J. and Singer, Y. (2004), An online algorithm for hierarchical phoneme classification, *in* 'Proceedings of the First International Conference on Machine Learning for Multimodal Interaction', MLMI'04, Springer-Verlag, Berlin, Heidelberg, pp. 146–158.

Garofolo, J. S., Lamel, L. F., Fisher, W. M., Fiscus, J. G., Pallett, D. S. and Dahlgren, N. L. (1993), 'TIMIT Acoustic-Phonetic Continuous Speech Corpus'.

Hamooni, H. and Mueen, A. (2014), Dual-Domain Hierarchical Classification of Phonetic Time Series, *in* '2014 IEEE International Conference on Data Mining', pp. 160–169.

Han, W., Pang, B. and Wu, Y. (2021), 'Robust Transfer Learning with Pretrained Language Models through Adapters'.

Hastie, T., Buja, A. and Tibshirani, R. (1995), 'Penalized Discriminant Analysis', *The Annals of Statistics* **23**(1), 73–102.

Hastie, T., Tibshirani, R. and Friedman, J. (2009), *The Elements of Statistical Learning*, Springer Series in Statistics, Springer, New York, NY.

Hayden, R. E. (1950), 'The Relative Frequency of Phonemes in General-American English', *WORD* **6**(3), 217–223.

He, J. and Chen, S. X. (2016), 'Testing super-diagonal structure in high dimensional covariance matrices', *Journal of Econometrics* **194**(2), 283–297.

Hu, G., Zhang, Y. and Yang, Q. (2019), Transfer Meets Hybrid: A Synthetic Approach for Cross-Domain Collaborative Filtering with Text, *in* 'The World Wide Web Conference', WWW '19, Association for Computing Machinery, New York, NY, USA, pp. 2822–2829.

Ke, Y., Minsker, S., Ren, Z., Sun, Q. and Zhou, W.-X. (2019), 'User-friendly covariance estimation for heavy-tailed distributions', *Statistical Science* **34**(3).

Lee, K.-F. and Hon, H.-W. (1989), 'Speaker-independent phone recognition using hidden Markov models', *IEEE Transactions on Acoustics, Speech, and Signal Processing* **37**(11), 1641–1648.

Lee, K., Lee, K. and Lee, J. (2022), 'Estimation of conditional mean operator under the bandable covariance structure', *Electronic Journal of Statistics* **16**(1), 1253–1302.

Li, S., Cai, T. T. and Li, H. (2022), 'Transfer Learning in Large-Scale Gaussian Graphical Models with False Discovery Rate Control', *Journal of the American Statistical Association* **0**(0), 1–13.

Li, S., Zhang, L., Cai, T. T. and Li, H. (2023), 'Estimation and Inference for High-Dimensional Generalized Linear Models with Knowledge Transfer', *Journal of the American Statistical Association* **0**(0), 1–12.

Li, Y., Ding, A. A. and Dy, J. (2017), Rate Optimal Estimation for High Dimensional Spatial Covariance Matrices, *in* 'Proceedings of the Ninth Asian Conference on Machine Learning', Vol. 77, PMLR, pp. 208–223.

Maqsood, M., Nazir, F., Khan, U., Aadil, F., Jamal, H., Mehmood, I. and Song, O.-y. (2019), 'Transfer Learning Assisted Classification and Detection of Alzheimer's Disease Stages Using 3D MRI Scans', *Sensors* **19**(11), 2645.

Minsker, S. and Wei, X. (2018), 'Estimation of the covariance structure of heavy-tailed distributions'.
URL: *http://arxiv.org/abs/1708.00502*

Petegrosso, R., Park, S., Hwang, T. H. and Kuang, R. (2017), 'Transfer learning across ontologies for phenome–genome association prediction', *Bioinformatics* **33**(4), 529–536.

Qiu, Y. and Chen, S. X. (2012), 'Test for bandedness of high-dimensional covariance matrices and bandwidth estimation', *The Annals of Statistics* **40**(3), 1285–1314.

Qiu, Y. and Chen, S. X. (2015), 'Bandwidth Selection for High-Dimensional Covariance Matrix Estimation', *Journal of the American Statistical Association* **110**(511), 1160–1174.

Raina, R., Ng, A. Y. and Koller, D. (2006), Constructing informative priors using transfer learning, *in* 'Proceedings of the 23rd International Conference on Machine Learning', ICML '06, Association for Computing Machinery, New York, NY, USA, pp. 713–720.

Rodrigues, P. L. C., Jutten, C. and Congedo, M. (2019), 'Riemannian Procrustes Analysis: Transfer Learning for Brain–Computer Interfaces', *IEEE Transactions on Biomedical Engineering* **66**(8), 2390–2401.

Vershynin, R. (2018), *High-Dimensional Probability: An Introduction with Applications in Data Science*, Cambridge Series in Statistical and Probabilistic Mathematics, Cambridge University Press, Cambridge.

Wainwright, M. J. (2019), *High-Dimensional Statistics: A Non-Asymptotic Viewpoint*, Cambridge Series in Statistical and Probabilistic Mathematics, Cambridge University Press, Cambridge.

Weiss, K., Khoshgoftaar, T. M. and Wang, D. (2016), 'A survey of transfer learning', *Journal of Big Data* **3**(1), 9.

Yi, F. and Zou, H. (2013), 'SURE-tuned tapering estimation of large covariance matrices', *Computational Statistics & Data Analysis* **58**, 339–351.

Zhuang, F., Qi, Z., Duan, K., Xi, D., Zhu, Y., Zhu, H., Xiong, H. and He, Q. (2021), 'A Comprehensive Survey on Transfer Learning', *Proceedings of the IEEE* **109**(1), 43–76.